

Narrative Report: Design and Implementation of Tracking Filters for a Passive Radar System

by Itumeleng Malemela

Overall Impression:

This is well-written dissertation that demonstrates strong specialist knowledge of the field in question and competent execution of research activities. The student mostly explains key concepts in a clear and concise manner that is largely accessible to non-specialist readers, with a few notable exceptions (to be discussed) where explanations could be improved or come sooner in the text. While it appears that the tests were conducted adeptly and thoroughly, there are some important aspects of the methodology that are not fully conveyed in this report, and the discussion of some results requires more context to be fully understandable. The dissertation is sorely in need of a summary of results across metrics and scenarios, and an overall conclusion regarding the best-performing filter method (perhaps with different priorities in mind) as the aim of designing a tracking filter does not feel complete without an explicit recommendation of which approach to follow.

My section-specific recommendations for improving this dissertation are as follows:

Introduction:

- 1) The introduction effectively establishes the background, problem statement, aims and scope, but, as a non-specialist in radar, there are some initial explanations of important concepts that should be improved.
 - Passive vs. traditional radar: the most critical aspect – that passive radar does not involve a dedicated transmitter, and instead uses transmission signals from external sources – does not clearly come across in this first description.
 - Surveillance and reference signals: state clearly that the reference signal is a reading of the external transmission signal, while the surveillance signal is the reflection from the target.
 - Direct path interference: rather than just stating that it is the transmission signal, it is important to mention that the interference is caused by this signal directly travelling to the receiver without bouncing off targets in between.
 - Constant false alarm rate (CFAR) map: give a brief one or two sentence description elaborating on what this is and how it is obtained. It is not obvious whether it is an algorithm, a probability distribution, a collection of measurements or some other form of information, or what its inputs and outputs are.
 - “More signal processing is required to process detections from the CFAR to tracks where targets can be visually followed, such as centroiding...” – the terminology that follows is not explained until much later in the dissertation, so it would be better to give more high-level descriptions e.g. instead of centroiding, “clustering of points into individual targets”, etc.
 - Bistatic range, doppler domain: these concepts are very important to defining the scope of the project, but they are not clearly and explicitly defined anywhere in the report. They should be explained somewhere early in the report.
- 2) Figure 1.1: The fonts on this plot are almost illegibly tiny. This is true of many figures in the dissertation, so the student should go through them all and ensure that no fonts in the

legends, labels or elsewhere have an effective size of smaller than 10 pt when resized and placed into the final document.

The ARD is a very specific visualization to this field, so it would be helpful to have more information on how to interpret this plot. What do the measurements on the axes mean in terms of the real-world movement or position of the target relative to the antennae? What are the important observable differences between the left and right plots and what do they mean?

- 3) “FM radio signals provide **low-range** resolution...” – this typo appears more than once, and it makes the meaning confusing. Is the range resolution low? Or is the resolution at low frequency ranges?

Literature Review:

The literature review is decently comprehensive and the approaches discussed are generally explained concisely and effectively brought back to the context of the project. Like the introduction, however, there are some initial explanations of ideas here that could be explained in slightly more detail for non-specialist readers.

- What is it that made the Kalman filter so revolutionary? Give a brief description of the technical feature(s) of this filter that differentiated it from approaches at the time. The Kalman filter subsection could also be improved by a more detailed explanation of the mechanism of this filter in the first paragraph, so its place in the space of methods and its limitations, can be better understood.
- The description of maximum likelihood estimation could be improved. Saying “ML maximizes the likelihood of the observations given the unknown parameters” does not make it clear that the unknown parameters are variables in an optimization problem. Something like “ML is an optimization approach that determines the values of the unknown parameters that maximize the likelihood of the observed data occurring” makes the mechanism more apparent.
- What were the “groundbreaking contributions of Norbert Wiener”? Give a brief, simple description of the key contributions, so it is clear how they set up the development of the Kalman filter.

Theory Development:

This is a very well-written section. The necessary technical information is explained effectively without excessive jargon. Equations are included sparingly, and fully annotated.

Simulations:

- 1) This is the bulk of the experimental work of the project. The tests are well-executed, but more explanation of some important aspects of the method is needed:
 - How was the target simulated? This is explained to some extent later in the report with the different aircraft movement scenarios, but it might be better to describe these test cases here and then identify which one is used in the preliminary parameter comparisons, as I was left wondering where the ground truth came from while reading.

- Likewise, what is the ground truth for the MTT tests? It was not clear to me what the correct paths of the various targets were supposed to look like, and therefore, whether they were reproduced accurately.
- The tests demonstrating the effects of different filter parameters were useful for establishing which parameters had an effect on the output, but it is difficult to quantify the sensitivity from looking at the plots. Summarizing the effects in terms of a cumulative error metric would help.
- It would be helpful to have a similar experiment for the parameters of the CA-CFAR (especially the P_{fa} parameter), as it is not explained how these values were selected. Particularly, a value of $P_{fa} = 1 \times 10^{-5}$ is listed in table 4.2, but figures 4.10 and 4.11 reference a value of $P_{fa} = 1 \times 10^{-7}$. Were these selected by trial and error? If so, how sensitive were the results to these changes?
- The same as the above can be said for the bandwidth parameter $\sigma = [5 \text{ Hz}, 10 \text{ km}]$ of the clustering algorithm. How was this selected and how sensitive were the results to this value?

2) Again, be careful of very small fonts on the figures (E.g. fig 4.13)

Performance Evaluation of Tracking Filters:

- 1) As in the previous section, the tests are well-executed and look at many aspects of the filters, but there is room for minor improvements in how these results are reported.
 - Figure 5.4, 5.5: It would be beneficial to have some additional context to interpret these results, namely, what is the threshold above which prediction time becomes unsuitable for real-time application? While it is very clear that the particle filter has a much longer prediction time, without the frame of reference of how long this time can be, it is unclear whether this difference is meaningful in a practical sense.
 - Data Association Performance: again – would be easier to interpret with more information about the reference targets for the MTT scenario.
 - Tables 5.7 to 5.9: it is not easy to pull out a clear result by just reading the numbers, so an additional visual stimulus would help to make these instantaneously interpretable. I would suggest highlighting the best performing result or results in each row with one colour, and the worst result or results with another, so you can see immediately which column tends to include more good or bad results.
- 2) It is critical to have an overarching summary of all results across the various metrics and scenarios. Something as simple as a colour-coded table, ranking the performance of all filters across the different tests performed e.g.

	Filter 1	Filter 2	Filter 3... etc.
Log likelihood, landing	1	2	3
Log likelihood, 360 manoeuvre	1	3	4
...			
Prediction time.. etc	3	1	2

would help to identify which filters performed best in particular contexts or metrics. This should then be used to make explicit recommendations about what the ultimate design should be, possibly with certain priorities in mind (e.g. what is the best approach if accuracy is critical? What is the best approach for minimizing load on hardware?)